

Problems Ch. 11

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ



1

1. The boiling point of CH_4 is much lower than that of HF . This is because:

A. of ion-dipole interactions in CH_4 .

B. of dipole-dipole interactions in CH_4

C. of hydrogen bonding in HF .

D. CH_4 is polar.

CH_4
↓
non polar
↓
London

HF
↓
polar
↓
- London
- dipole-dipole
- H-bonding

B.P ↑
I.M.F ↑

2

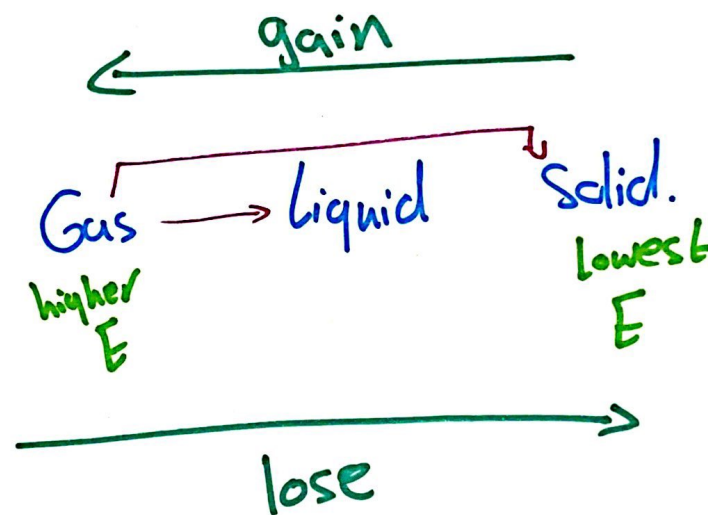
2. For a given substance, which of the following phase transitions should RELEASE the most energy

A. Liquid to gas

☒ B. Gas to solid

C. Gas to liquid

D. Solid to gas

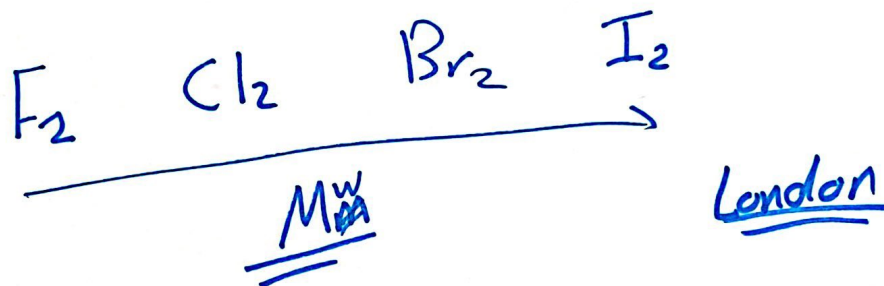


3

3. At room temperature, F_2 and Cl_2 are gases, Br_2 is a liquid, and I_2 is a solid. This is because:

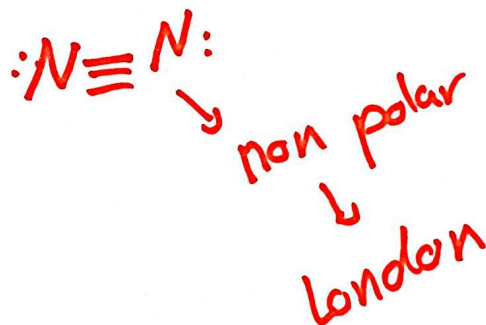
- A. dipole-induced dipole interactions increase with molecular size.
- B. polarity increases with molecular size.
- C. dipole-dipole interactions increase with molecular size.
- ☒ D. dispersion interactions increase with molecular size.

London



What type of intermolecular forces holds liquid N₂ together?

- A. ionic bonding **B. London forces** C. hydrogen bonding D. dipole-dipole interaction



Which response includes only those compounds that can exhibit hydrogen bonding?

A. AsH_3 , H_2Te

X

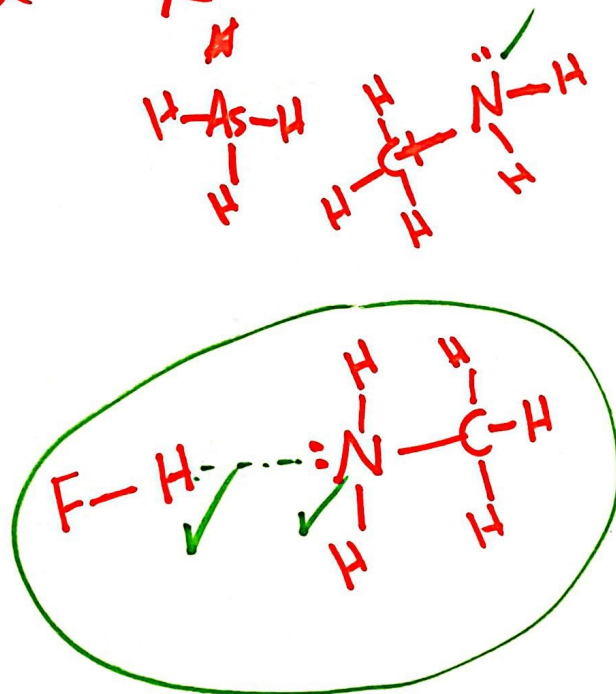
B. AsH_3 , CH_3NH_2

X

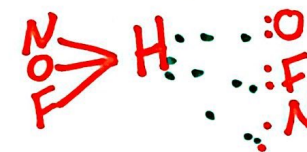
C. CH_4 , AsH_3 , H_2Te

X

☒ D. CH_3NH_2 , HF



H bond



Which of the following boils at the highest temperature?

- A. CH_4 B. C_2H_6 C. C_3H_8 D. C_4H_{10}

London $\uparrow$
size $\uparrow$

B.P $\uparrow$
M.F $\uparrow$

6

7

Which probably has the lowest boiling point at 1.00 atm?

A. ~~HF~~

B. HCl

C. HBr

D. HI

↓
polar
↓
London
dipole-d
H-bond

↓
polar
↓
London
d-d.

↓
polar
↓
London
d-d

↓
polar
↓
London
d-d.

IMF ↓

NOF



The normal boiling point of a liquid is

- ☒ .A. the temperature at which the vapor pressure equals 760 torr
- .B. the temperature above which the substance cannot exist as a liquid regardless of the pressure
- .C. the only temperature at which there can be equilibrium between liquid and gas
- .D. the temperature at which the liquid will usually boil

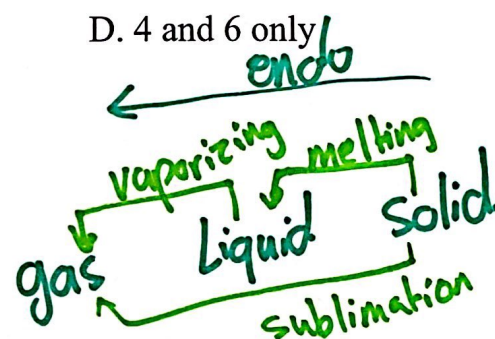
Which of the following phase changes is(are) endothermic?

1. melting 3. sublimation 5. Deposition

2. vaporization 4. condensation 6. freezing

A. 1, 2, and 3 B. 4, 5, and 6 C. 1 and 2 only

D. 4 and 6 only



10

Which one of the following elements is least likely to participate in a hydrogen bond?

A. O

B. F

C. S

D. N

N O F

Which of the following compounds has the highest boiling point?

A. CH_4

↓
London

B. CH_3CH_3

↓
London

C. $\text{CH}_3\text{CH}_2\text{OH}$

↓
London
dipole-dipole
H-bond.

D. CH_3OCH_3

↓
London
d-d.

B.P ↑
IMF ↑

11

12

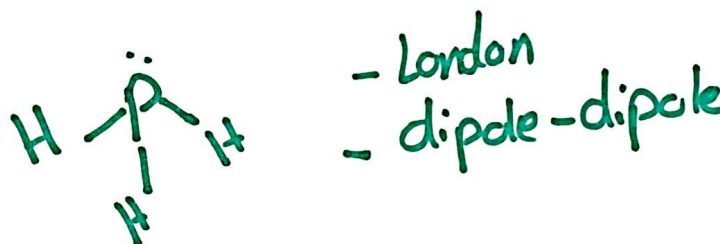
Which of the following best describes all the intermolecular forces exhibited by a pure sample of PH₃?

A. dispersion, dipole-dipole, and hydrogen bonding

B. dispersion only

C. dispersion and hydrogen bonding

☒ D. dispersion and dipole-dipole

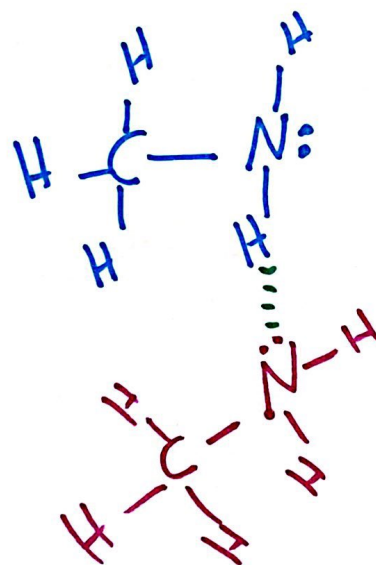


Which of the following best describes all the intermolecular forces exhibited by a pure sample of CH_3NH_2 ?

- ☒ A. dispersion, dipole-dipole, and hydrogen bonding
C. dispersion and hydrogen bonding

B. dispersion only

D. dispersion and dipole-dipole



- London
- d-d
- H-bond

Which of the following best describes all the intermolecular forces exhibited by a pure sample of CS₂?

- A. dispersion, dipole-dipole, and hydrogen bonding
- C. dispersion and hydrogen bonding

- ☒ B. dispersion only
- D. dispersion and dipole-dipole



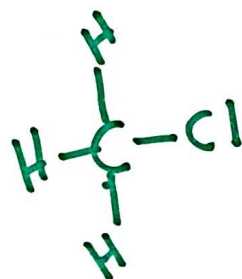
Which of the following best describes all the intermolecular forces exhibited by a pure sample of CH_3Cl ?

A. dispersion, dipole-dipole, and hydrogen bonding

B. dispersion only

C. dispersion and hydrogen bonding

☒ D. dispersion and dipole-dipole



London
d-d

Which molecule would be expected to have the highest viscosity?



↓
London
dipole-d
2 H-bond



↓
London
d-d



↓
London
d-d
H-bond



↓
London
d-d
H-b

Viscosity ↑
IMF ↑
Mw ↑
temp ↓

Which pure substance would have the smallest molar heat of vaporization?

A. C_4H_{10}

B. C_5H_{12}

C. C_3H_8

D. CH_4

↑
IMF ↑

17

Which molecule would be expected to have the ¹⁸ highest vapor pressure?

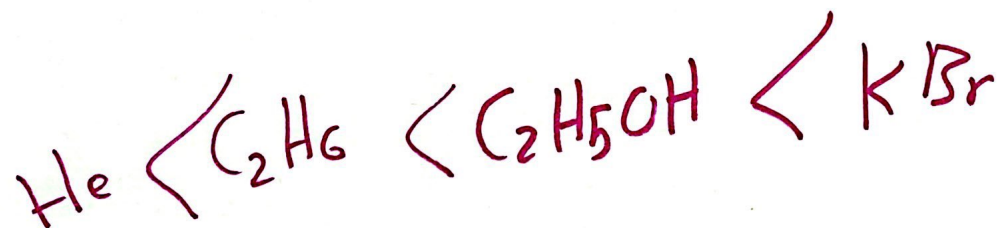
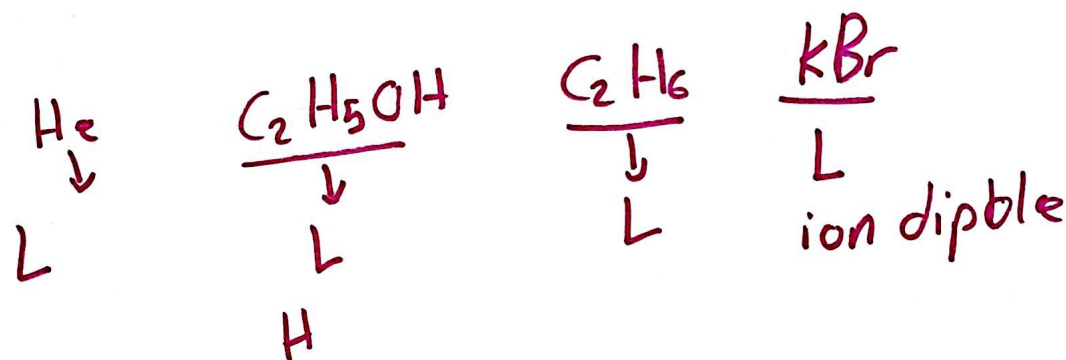


vapor pressure $\uparrow$
IMF $\downarrow$



The correct order of increasing boiling point is

- A. $\text{He} < \text{C}_2\text{H}_5\text{OH} < \text{C}_2\text{H}_6 < \text{KBr}$ B. $\text{KBr} < \text{C}_2\text{H}_6 < \text{C}_2\text{H}_5\text{OH} < \text{He}$
 C. $\text{He} < \text{C}_2\text{H}_6 < \text{C}_2\text{H}_5\text{OH} < \text{KBr}$ D. $\text{C}_2\text{H}_5\text{OH} < \text{C}_2\text{H}_6 < \text{He} < \text{KBr}$



For a pure substance, the $\Delta H_{\text{freezing}}$ and $\Delta H_{\text{deposition}}$ are known to be -15 kJ/mol and -75 kJ/mol

respectively. What is $\Delta H_{\text{vaporization}}$ for the substance

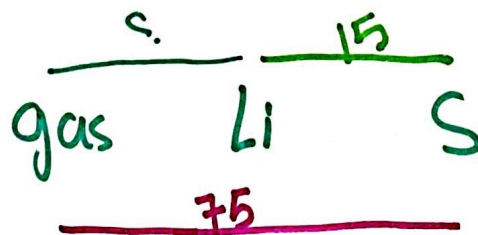
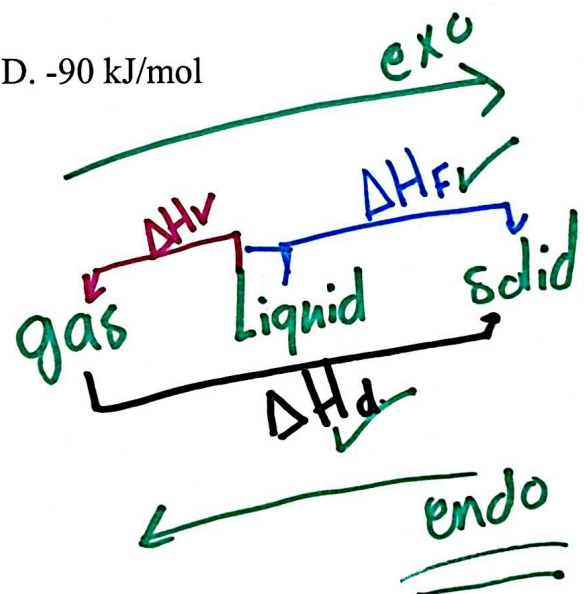
A. 60 kJ/mol

B. -60 kJ/mol

C. 90 kJ/mol

D. -90 kJ/mol

60 kJ/mol



IMF ↑

The correct order of the viscosity is -----, All are at the same 50°C temperature

I. $\text{CH}_3\text{CH}_2\text{OCH}_3$ II. $\text{CH}_3(\text{CH}_2)_2\text{NH}_2$ III. $\text{CH}_3(\text{CH}_2)_4\text{NH}_2$ IV. $\text{CH}_3(\text{CH}_2)_2\text{CH}_3$

A. $\text{I} > \text{II} > \text{III} > \text{IV}$ (B.) $\text{III} > \text{II} > \text{I} > \text{IV}$ C. $\text{IV} > \text{II} > \text{I} > \text{III}$ D. $\text{IV} > \text{I} > \text{II} > \text{III}$

L
d-d

L
d-d
H-bond

L
d-d
H-bond.

L

$\text{III} > \text{II} > \text{I} > \text{IV}$

IMF ↑

H₂O highest

22

The correct order of the surface tension is ----- . All are at the same 50°C temperature

I. CH3CH2OCH3 II. CH3(CH2)2NH2 III. CH3(CH2)4NH2 IV. CH3(CH2)2CH3

A. I > II > III > IV B. III > II > I > IV C. IV > II > I > III D. IV > I > II > III

Which of the following liquids would have the lowest viscosity, factoring in both the impact of the substance and the temperature?

A. $\text{C}_3\text{H}_7\text{OH}$ at 25°C

L
d-d
H

B. $\text{C}_3\text{H}_7\text{OH}$ at 75°C

L
d-d
H

C. MgBr_2 at 25°C

↓
ion-dipole

D. $\text{C}_5\text{H}_{11}\text{OH}$ at 75°C

↓
L
d-d
H

Viscosity ↓
IMF ↓
temp ↑

25
24

Predict which of the following liquid/temperature scenarios would have the HIGHEST vapor pressure and the LOWEST surface tension?

A. C_6H_{14} at 275 K

↓
↓

B. C_6H_{14} at 299 K

↓
↓

C. C_5H_{12} at 299 K

↓
↓

D. HOC_4H_8OH at 299 K

↓
2H

surface tension ↓
(IMF ↓)

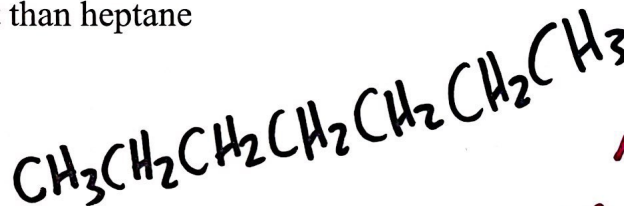
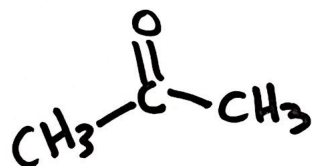
vapor pressure ↑
IMF ↓

IMF $\rightarrow$ acetone $<$ heptane $<$ ethanol

25

At room temperature, the vapor pressure pattern is acetone $>$ heptane $>$ ethanol. Which one of the following statements is FALSE?

- ☒ A. a substance with higher vapor pressure is held together by weaker binding forces
- ☒ B. ethanol has the lowest vapor pressure and strongest intermolecular force due to hydrogen bonding
- ☒ C. heptane has lower vapor pressure than acetone due to London dispersion forces
- ☐ D. acetone would have a higher boiling point than heptane



hexane $\uparrow$

$\checkmark$ $\rightarrow$